

# Waypoint vs. Gauss-Markov Mobility Comparison: (Phase 2 Item 1, Revised)

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- **Total simulation runs evaluated:** 648
- **Revision note:** this supersedes the June submission. Per reviewer feedback: seeds expanded from 1 to 3 (7/8/9) with real mean/SD statistics instead of " $\pm 0.00\%$ "; RSSI threshold standardized to the project's  $-58$  dBm (was  $-80$  dBm); the "work" scenario's building-site placement was rebalanced so it no longer over-samples the field's worst-covered corners; results are broken out per robot type instead of blended into one "waypoint average"; STA count, payload, and cellular mode are now swept (previously fixed); and every run now has a 3D trajectory plot, animation, RSSI heatmap, and switching timeline.

## Experimental design overview

Each configuration corresponds to one simulation run comparing the Waypoint + dwell-time mobility model (patrol / transport / work robot archetypes) against the existing GaussMarkovMobilityModel baseline, now crossed with STA count, payload size, and cellular mode -- the same sweep axes the original Phase 1 matrix used -- so results are directly comparable to it.

| Design element       | Specification                                 |
|----------------------|-----------------------------------------------|
| Total configurations | 648 simulation runs                           |
| Scenarios            | gaussmarkov_baseline, patrol, transport, work |
| Speeds swept         | 0.5, 2.0, 5.0 m/s                             |
| STA counts swept     | 5, 10, 15                                     |
| Payloads swept       | 10kb, 50kb, 1mb                               |
| Cellular modes swept | LTE, NR                                       |

| Design element                                       | Specification                                                                            |
|------------------------------------------------------|------------------------------------------------------------------------------------------|
| RNG seeds                                            | 7, 8, 9 (project convention: 3-repetition average)                                       |
| Parameters held constant                             | hotspotBand=5g; meshConfig=1; simulation duration 90s; RSSI handover threshold -58 dBm   |
| Canonical config for headline/speed/verdict sections | LTE, 5 STA, 1mb payload, 5.0 m/s (see Sections 3-5 for STA/payload/cellular sensitivity) |

## 1. Head-to-Head Comparison @ Canonical Config (LTE, 5 STA, 1mb, 5.0 m/s)

This section presents the top-level switching and signal-quality indicators for each mobility scenario at one fixed configuration, each now averaged over 3 seeds with real mean  $\pm$  SD (95% CI in parentheses where shown) -- not the single-run " $\pm 0.00\%$ " from the June submission.

### Section Summary

- **Best reliability:** `transport` reaches  **$98.96 \pm 0.95$  (n=3)%** packet-weighted PDR.
- **Fastest switch recovery:** `work` averages  **$15.1 \pm 22.2$  (n=3) ms** service interruption.
- **Slowest switch recovery:** `gaussmarkov_baseline` averages  **$32.2 \pm 32.4$  (n=3) ms** service interruption.
- **Gauss-Markov RSSI abruptness:** p95  $|\Delta\text{RSSI}| = \mathbf{2.71 \pm 1.43}$  (n=3) dB -- the reference point Section 6 checks against.

| Scenario             | Runs | Switch events / 100s     | Burstiness (CoV)         | Mean $ \Delta\text{RSSI} $ (dB) | p95 $ \Delta\text{RSSI} $ (dB) | Mean interruption (ms)   | Packet-weighted PDR (%)   |
|----------------------|------|--------------------------|--------------------------|---------------------------------|--------------------------------|--------------------------|---------------------------|
| gaussmarkov_baseline | 3    | $4.81 \pm 0.64$<br>(n=3) | $3.00 \pm 0.00$<br>(n=3) | $0.80 \pm 0.21$<br>(n=3)        | $2.71 \pm 1.43$<br>(n=3)       | $32.2 \pm 32.4$<br>(n=3) | $94.69 \pm 4.55$<br>(n=3) |
| patrol               | 3    | $4.44 \pm 1.11$<br>(n=3) | $2.51 \pm 0.44$<br>(n=3) | $0.33 \pm 0.03$<br>(n=3)        | $0.58 \pm 0.02$<br>(n=3)       | $28.4 \pm 29.1$<br>(n=3) | $98.51 \pm 1.55$<br>(n=3) |

| Scenario  | Runs | Switch events / 100s     | Burstiness (CoV)         | Mean   $\Delta$ RSSI  (dB) | p95   $\Delta$ RSSI  (dB) | Mean interruption (ms)   | Packet-weighted PDR (%)   |
|-----------|------|--------------------------|--------------------------|----------------------------|---------------------------|--------------------------|---------------------------|
| transport | 3    | 5.56 $\pm$ 0.00<br>(n=3) | 1.95 $\pm$ 0.41<br>(n=3) | 0.44 $\pm$ 0.02<br>(n=3)   | 1.44 $\pm$ 0.03<br>(n=3)  | 30.0 $\pm$ 50.3<br>(n=3) | 98.96 $\pm$ 0.95<br>(n=3) |
| work      | 3    | 5.19 $\pm$ 0.64<br>(n=3) | 3.00 $\pm$ 0.00<br>(n=3) | 0.13 $\pm$ 0.03<br>(n=3)   | 0.41 $\pm$ 0.20<br>(n=3)  | 15.1 $\pm$ 22.2<br>(n=3) | 92.51 $\pm$ 3.17<br>(n=3) |

95% confidence intervals (t-distribution, n=3 -- wide by construction; a firmer CI needs enhancement item 4's planned 10-seed expansion):

| Scenario             | Switch events / 100s 95% CI | Mean interruption (ms) 95% CI | PDR (%) 95% CI  |
|----------------------|-----------------------------|-------------------------------|-----------------|
| gaussmarkov_baseline | [3.22, 6.41]                | [-48.3, 112.8]                | [83.39, 105.99] |
| patrol               | [1.68, 7.20]                | [-43.9, 100.7]                | [94.65, 102.37] |
| transport            | [5.56, 5.56]                | [-94.9, 154.9]                | [96.61, 101.32] |
| work                 | [3.59, 6.78]                | [-40.1, 70.3]                 | [84.64, 100.39] |

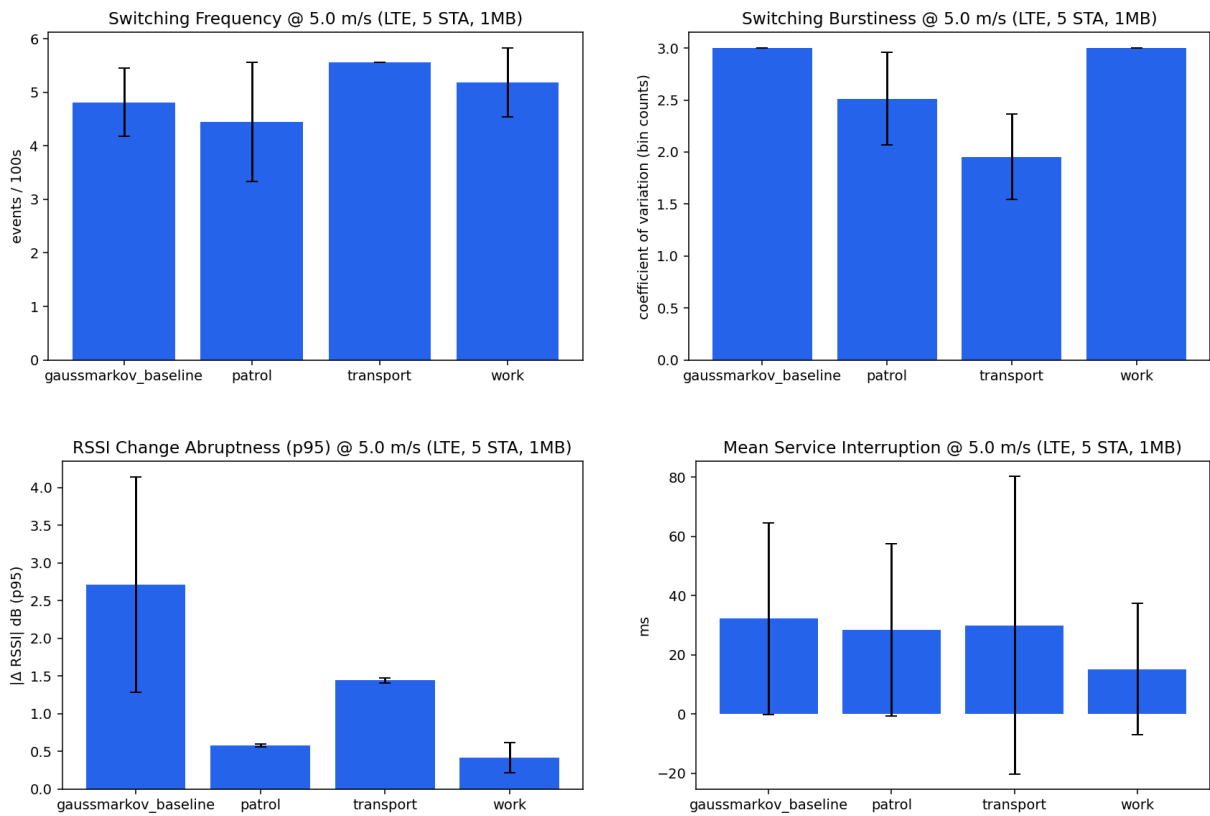


Figure: Switching frequency, burstiness, RSSI-change abruptness, and mean service interruption across scenarios at the canonical config.

## 2. Waypoint Speed Sensitivity (0.5 / 2.0 / 5.0 m/s)

Canonical cellular/STA/payload held fixed (LTE, 5 STA, 1mb); speed is the variable of interest, per the enhancement plan's 3-level speed comparison (section 2.3).

### Section Summary

- patrol from **0.5** to **5.0 m/s**: p95  $|\Delta \text{RSSI}|$  0.06  $\rightarrow$  0.58 dB (+0.51 dB).
- transport from **0.5** to **5.0 m/s**: p95  $|\Delta \text{RSSI}|$  0.37  $\rightarrow$  1.44 dB (+1.07 dB).
- work from **0.5** to **5.0 m/s**: p95  $|\Delta \text{RSSI}|$  0.06  $\rightarrow$  0.41 dB (+0.35 dB).

| Scenario | Speed (m/s) | Runs | Switch events / 100s  | Burstiness (CoV)      | Mean $ \Delta \text{RSSI} $ (dB) | p95 $ \Delta \text{RSSI} $ (dB) |
|----------|-------------|------|-----------------------|-----------------------|----------------------------------|---------------------------------|
| patrol   | 0.5         | 3    | $3.70 \pm 0.64$ (n=3) | $3.00 \pm 0.00$ (n=3) | $0.17 \pm 0.04$ (n=3)            | $0.06 \pm 0.00$ (n=3)           |
| patrol   | 2.0         | 3    | $4.44 \pm 0.00$ (n=3) | $2.53 \pm 0.41$ (n=3) | $0.23 \pm 0.04$ (n=3)            | $0.24 \pm 0.01$ (n=3)           |

| Scenario  | Speed (m/s) | Runs | Switch events / 100s  | Burstiness (CoV)      | Mean   $\Delta$ RSSI  (dB) | p95   $\Delta$ RSSI  (dB) |
|-----------|-------------|------|-----------------------|-----------------------|----------------------------|---------------------------|
| patrol    | 5.0         | 3    | 4.44 $\pm$ 1.11 (n=3) | 2.51 $\pm$ 0.44 (n=3) | 0.33 $\pm$ 0.03 (n=3)      | 0.58 $\pm$ 0.02 (n=3)     |
| transport | 0.5         | 3    | 3.70 $\pm$ 0.64 (n=3) | 3.00 $\pm$ 0.00 (n=3) | 0.10 $\pm$ 0.03 (n=3)      | 0.37 $\pm$ 0.01 (n=3)     |
| transport | 2.0         | 3    | 4.81 $\pm$ 0.64 (n=3) | 1.93 $\pm$ 0.33 (n=3) | 0.25 $\pm$ 0.03 (n=3)      | 0.80 $\pm$ 0.05 (n=3)     |
| transport | 5.0         | 3    | 5.56 $\pm$ 0.00 (n=3) | 1.95 $\pm$ 0.41 (n=3) | 0.44 $\pm$ 0.02 (n=3)      | 1.44 $\pm$ 0.03 (n=3)     |
| work      | 0.5         | 3    | 5.19 $\pm$ 0.64 (n=3) | 3.00 $\pm$ 0.00 (n=3) | 0.06 $\pm$ 0.04 (n=3)      | 0.06 $\pm$ 0.03 (n=3)     |
| work      | 2.0         | 3    | 5.19 $\pm$ 0.64 (n=3) | 3.00 $\pm$ 0.00 (n=3) | 0.09 $\pm$ 0.02 (n=3)      | 0.24 $\pm$ 0.13 (n=3)     |
| work      | 5.0         | 3    | 5.19 $\pm$ 0.64 (n=3) | 3.00 $\pm$ 0.00 (n=3) | 0.13 $\pm$ 0.03 (n=3)      | 0.41 $\pm$ 0.20 (n=3)     |

Figure interpretation: RSSI-change abruptness increases with speed across all three robot types -- faster movement covers more distance between fixed-interval samples, so signal strength changes more per second.

### 3. STA Count Sensitivity (5 / 10 / 15 STAs)

Canonical cellular/payload/speed held fixed (LTE, 1mb, 5.0 m/s); STA count is swept, matching the original Phase 1 matrix's granularity for this axis. This is new in the July revision -- June fixed STA count at 5 throughout.

| Scenario             | STA count | Runs | Switch events / 100s   | Mean interruption (ms) | PDR (%)                |
|----------------------|-----------|------|------------------------|------------------------|------------------------|
| gaussmarkov_baseline | 5         | 3    | 4.81 $\pm$ 0.64 (n=3)  | 32.2 $\pm$ 32.4 (n=3)  | 94.69 $\pm$ 4.55 (n=3) |
| gaussmarkov_baseline | 10        | 3    | 10.74 $\pm$ 0.64 (n=3) | 2.8 $\pm$ 0.9 (n=3)    | 92.53 $\pm$ 2.22 (n=3) |
| gaussmarkov_baseline | 15        | 3    | 16.67 $\pm$ 0.00 (n=3) | 14.5 $\pm$ 10.7 (n=3)  | 89.03 $\pm$ 7.83 (n=3) |

| Scenario  | STA count | Runs | Switch events / 100s   | Mean interruption (ms) | PDR (%)                 |
|-----------|-----------|------|------------------------|------------------------|-------------------------|
| patrol    | 5         | 3    | $4.44 \pm 1.11$ (n=3)  | $28.4 \pm 29.1$ (n=3)  | $98.51 \pm 1.55$ (n=3)  |
| patrol    | 10        | 3    | $8.52 \pm 0.64$ (n=3)  | $80.9 \pm 28.0$ (n=3)  | $96.67 \pm 1.00$ (n=3)  |
| patrol    | 15        | 3    | $13.70 \pm 1.70$ (n=3) | $37.0 \pm 30.7$ (n=3)  | $82.85 \pm 17.59$ (n=3) |
| transport | 5         | 3    | $5.56 \pm 0.00$ (n=3)  | $30.0 \pm 50.3$ (n=3)  | $98.96 \pm 0.95$ (n=3)  |
| transport | 10        | 3    | $10.74 \pm 0.64$ (n=3) | $37.0 \pm 32.8$ (n=3)  | $96.27 \pm 4.76$ (n=3)  |
| transport | 15        | 3    | $15.56 \pm 1.11$ (n=3) | $21.2 \pm 21.0$ (n=3)  | $91.82 \pm 4.32$ (n=3)  |
| work      | 5         | 3    | $5.19 \pm 0.64$ (n=3)  | $15.1 \pm 22.2$ (n=3)  | $92.51 \pm 3.17$ (n=3)  |
| work      | 10        | 3    | $9.63 \pm 0.64$ (n=3)  | $46.1 \pm 45.6$ (n=3)  | $94.79 \pm 1.60$ (n=3)  |
| work      | 15        | 3    | $15.56 \pm 1.11$ (n=3) | $58.4 \pm 43.9$ (n=3)  | $93.98 \pm 1.58$ (n=3)  |

## 4. Payload Sensitivity (10KB / 50KB / 1MB)

Canonical cellular/STA/speed held fixed (LTE, 5 STA, 5.0 m/s); payload size is swept. New in the July revision.

| Scenario                 | Payload | Runs | Switch events / 100s  | Mean interruption (ms)    | PDR (%)                |
|--------------------------|---------|------|-----------------------|---------------------------|------------------------|
| gaussmarkov_<br>baseline | 10kb    | 3    | $4.81 \pm 0.64$ (n=3) | $1974.7 \pm 2356.0$ (n=2) | $99.14 \pm 0.77$ (n=3) |
| gaussmarkov_<br>baseline | 50kb    | 3    | $4.81 \pm 0.64$ (n=3) | $1963.2 \pm 2351.2$ (n=2) | $99.14 \pm 0.73$ (n=3) |
| gaussmarkov_<br>baseline | 1mb     | 3    | $4.81 \pm 0.64$ (n=3) | $32.2 \pm 32.4$ (n=3)     | $94.69 \pm 4.55$ (n=3) |

| Scenario  | Payload | Runs | Switch events / 100s     | Mean interruption (ms)    | PDR (%)                    |
|-----------|---------|------|--------------------------|---------------------------|----------------------------|
| patrol    | 10kb    | 3    | $2.96 \pm 0.64$<br>(n=3) | $302.9 \pm 20.0$<br>(n=2) | $100.00 \pm 0.01$<br>(n=3) |
| patrol    | 50kb    | 3    | $2.96 \pm 0.64$<br>(n=3) | $275.0 \pm 52.4$<br>(n=2) | $99.98 \pm 0.03$<br>(n=3)  |
| patrol    | 1mb     | 3    | $4.44 \pm 1.11$<br>(n=3) | $28.4 \pm 29.1$<br>(n=3)  | $98.51 \pm 1.55$<br>(n=3)  |
| transport | 10kb    | 3    | $5.56 \pm 0.00$<br>(n=3) | $299.2 \pm 0.0$<br>(n=1)  | $99.99 \pm 0.02$<br>(n=3)  |
| transport | 50kb    | 3    | $5.56 \pm 0.00$<br>(n=3) | $297.6 \pm 0.0$<br>(n=1)  | $99.98 \pm 0.03$<br>(n=3)  |
| transport | 1mb     | 3    | $5.56 \pm 0.00$<br>(n=3) | $30.0 \pm 50.3$<br>(n=3)  | $98.96 \pm 0.95$<br>(n=3)  |
| work      | 10kb    | 3    | $2.96 \pm 2.31$<br>(n=3) | $311.4 \pm 0.0$<br>(n=1)  | $99.55 \pm 0.79$<br>(n=3)  |
| work      | 50kb    | 3    | $2.96 \pm 2.31$<br>(n=3) | $308.3 \pm 0.0$<br>(n=1)  | $99.50 \pm 0.82$<br>(n=3)  |
| work      | 1mb     | 3    | $5.19 \pm 0.64$<br>(n=3) | $15.1 \pm 22.2$<br>(n=3)  | $92.51 \pm 3.17$<br>(n=3)  |

**Caveat -- read the interruption column with this in mind:** at 10KB/50KB payload, interruption times are higher than at 1MB, which looks backwards. Checking the raw switch logs shows why: at small payloads the TCP flow finishes transferring almost immediately, so by the time a WiFi->cellular switch happens there is often no more application traffic in flight. "Time to first RX after switch" then measures how long until the next packet happens to be generated (sometimes never, hence some runs logging `timeout` status with multi-second "durations") rather than genuine network path-recovery speed. At 1MB the flow is still actively transferring, so recovery is detected within milliseconds. This is a property of the interruption metric's definition (last-good-RX to first-RX-after-switch), not a real payload-dependent slowdown in the switching mechanism itself -- and it is also why several payload rows above show n=1 or n=2 instead of n=3: some seeds had zero `resolved` switch events to average at all, only timeouts.

## 5. Cellular Mode Comparison (LTE vs. NR)

Canonical STA/payload/speed held fixed (5 STA, 1mb, 5.0 m/s); cellular fallback mode is swept. New in the July revision -- June only tested LTE.

| Scenario             | Cellular | Runs | Switch events / 100s  | Mean interruption (ms) | PDR (%)                |
|----------------------|----------|------|-----------------------|------------------------|------------------------|
| gaussmarkov_baseline | LTE      | 3    | $4.81 \pm 0.64$ (n=3) | $32.2 \pm 32.4$ (n=3)  | $94.69 \pm 4.55$ (n=3) |
| gaussmarkov_baseline | NR       | 3    | $5.19 \pm 0.64$ (n=3) | $19.1 \pm 22.3$ (n=3)  | $93.61 \pm 4.26$ (n=3) |
| patrol               | LTE      | 3    | $4.44 \pm 1.11$ (n=3) | $28.4 \pm 29.1$ (n=3)  | $98.51 \pm 1.55$ (n=3) |
| patrol               | NR       | 3    | $4.44 \pm 1.11$ (n=3) | $27.5 \pm 26.6$ (n=3)  | $97.17 \pm 3.48$ (n=3) |
| transport            | LTE      | 3    | $5.56 \pm 0.00$ (n=3) | $30.0 \pm 50.3$ (n=3)  | $98.96 \pm 0.95$ (n=3) |
| transport            | NR       | 3    | $5.56 \pm 0.00$ (n=3) | $30.6 \pm 50.0$ (n=3)  | $98.95 \pm 0.68$ (n=3) |
| work                 | LTE      | 3    | $5.19 \pm 0.64$ (n=3) | $15.1 \pm 22.2$ (n=3)  | $92.51 \pm 3.17$ (n=3) |
| work                 | NR       | 3    | $5.19 \pm 0.64$ (n=3) | $22.2 \pm 33.3$ (n=3)  | $97.09 \pm 0.23$ (n=3) |

## 6. Verdict Against the Enhancement Plan's Predictions

The enhancement plan predicts two effects of switching from Gauss-Markov to Waypoint mobility: switching events should become more concentrated (vs. dispersed), and RSSI variation should become more abrupt in specific zones (vs. gradual). **Each robot type is checked individually against the Gauss-Markov baseline below** -- the June report blended patrol/transport/work into one "waypoint average" here, which masked how differently "work" behaves from the other two (reviewer finding).

## Section Summary

**patrol** vs. Gauss-Markov:

- Switching dispersed → concentrated: NOT CONFIRMED -- burstiness 3.00 → 2.51 (baseline already at the metric's ceiling -- see caveat below).
- RSSI gradual → abrupt-change zones: NOT CONFIRMED -- p95  $|\Delta\text{RSSI}|$  2.71 → 0.58 dB.

**transport** vs. Gauss-Markov:

- Switching dispersed → concentrated: NOT CONFIRMED -- burstiness 3.00 → 1.95 (baseline already at the metric's ceiling -- see caveat below).
- RSSI gradual → abrupt-change zones: NOT CONFIRMED -- p95  $|\Delta\text{RSSI}|$  2.71 → 1.44 dB.

**work** vs. Gauss-Markov:

- Switching dispersed → concentrated: NOT CONFIRMED -- burstiness 3.00 → 3.00 (baseline already at the metric's ceiling -- see caveat below).
- RSSI gradual → abrupt-change zones: NOT CONFIRMED -- p95  $|\Delta\text{RSSI}|$  2.71 → 0.41 dB.

**Reading these as a preliminary signal, not a confirmed result:** each comparison above is  $n=3$  seeds per side. That is enough to move past June's single-run " $\pm 0.00\%$ " problem, but not enough for the statistical confidence enhancement item 4 targets (10 seeds, bootstrap 95% CIs, Mann-Whitney U / Kruskal-Wallis significance tests, Sep 2026). Treat "CONFIRMED (preliminary)" above as this data points that way, not as a statistically significant finding.

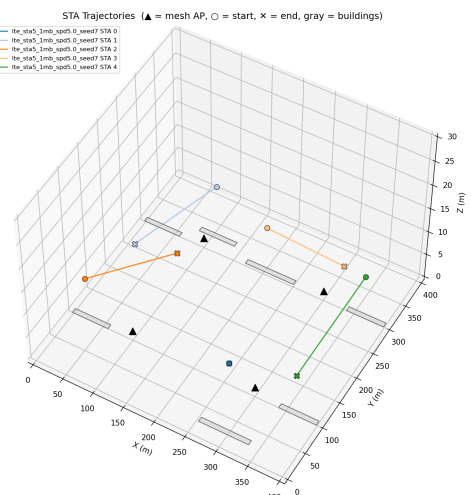
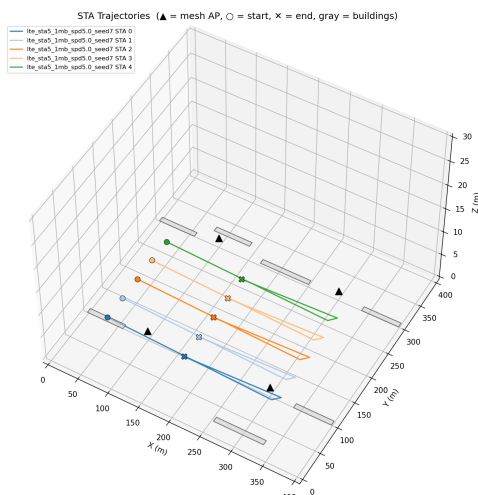
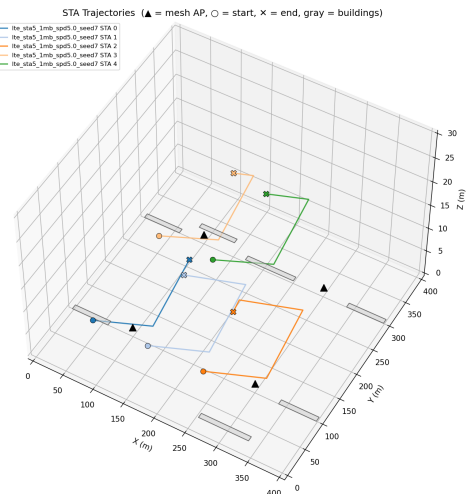
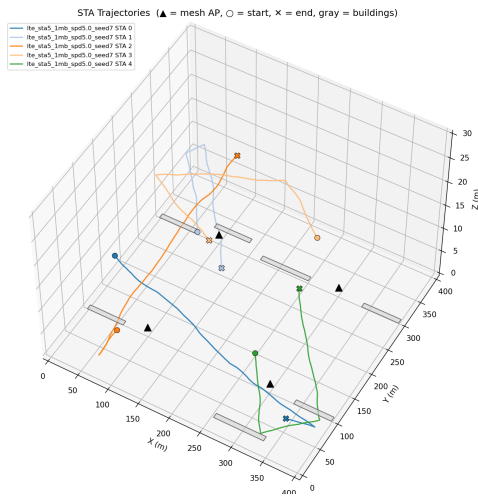
**Caveat on the burstiness comparison specifically:** at this canonical config, total switch events per run are low (roughly 4-5 events over 90s), and checking the raw switch logs shows they almost all land within the same  $\sim 9$ s time bin right at simulation start ( $t \approx 10$ -12s, when the WiFi RSSI-averaging window first fills and STAs make their first path decision) regardless of mobility model. That pins the Gauss-Markov baseline's burstiness at the metric's mathematical ceiling (3.00) before mobility is even a factor, so "dispersed → concentrated" has no room to show improvement at this operating point -- the metric isn't wrong, it's just saturated here. The RSSI-abruptness comparison above doesn't have this ceiling problem and is the more trustworthy of the two checks at this config.

## 7. Visual Deliverables

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Every one of the 648 runs in this sweep has its own 3D trajectory plot, node-movement animation, RSSI heatmap, and switching timeline GIF (files `trajectory_3d.png`, `animation.gif`, `rss_i_heatmap.png`,

switching\_timeline.gif, trajectory\_viewer.html in each run's output directory)  
 -- these were the deliverables missing from the June submission. One representative 3D trajectory per scenario type, at the canonical config, is embedded below; the full set lives on disk per-run rather than being bundled into this report.



Data note: 8 run(s) had zero switch events (all `work` scenario, low STA count/speed, seed 9) -- their `switching_timeline.gif` legitimately has nothing to animate; all other outputs for those runs are intact.